

Multiple Choice (10 marks)

1. Are Python variable names case-sensitive?
 - (a) Yes
 - (b) No
 - (c) It's machine-dependent
 - (d) None of the above
2. What is the value of this python expression: $1 + 3 \% 3$?
 - (a) 0
 - (b) 1
 - (c) 3
 - (d) 4
3. In Python 3, what is the output of `print tuple[0]` if `tuple = ( 'abcd', 786 , 2.23, 'john', 70.2 )`?
 - (a) ('abcd', 786 , 2.23, 'john', 70.2)
 - (b) abcd
 - (c) Error
 - (d) None of the above.
4. Which of the following best explains what happens when a new device is connected to the Internet?
 - (a) A device driver is assigned to the device.
 - (b) An Internet Protocol (IP) address is assigned to the device.
 - (c) A packet number is assigned to the device.
 - (d) A Web site is assigned to the device.
5. What is the output of `4*1**4` in python?
 - (a) 4
 - (b) 1
 - (c) 256
 - (d) 16

True / False (10 marks)

Answer True or False

6. True or False: In Python 3, the `seed([x])` function sets the starting value for generating random numbers.
7. True or False: The condition `a < c` is sufficient to ensure the expression evaluates to true for integers `a`, `b`, and `c`.
8. True or False: In Python 3, the `max` function returns the largest value from a list, so `max([1, 2, 3, 4])` is 4.
9. True or False: In Python 3, the `strip()` function removes all leading and trailing whitespace from a string.
10. True or False: In Python 3, the expression `x << 3` results in 16 when `x` is equal to 1.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the role of a Web server in the Internet ecosystem, explaining how it functions to deliver Web pages to clients and its importance in web communication.
12. Discuss how the selection sort algorithm operates in terms of data movement and comparisons. Why is the number of comparisons not affected by the initial arrangement of elements?

End of Paper